

Project Synopsis

On

Online Voting Survey and Election Opinion Analysis System

Submitted by

Udhayarajan J

MCA I (Semester: I)

**Guided by
Monica Mali**

Assistant Professor, MCA Department



**P.E. Society's
Modern Institute of Business Studies (Autonomous),
Nigdi, Pune - 411044**

2025 - 2026

❖ **Student Name:** Udhayarajan J

❖ **Title Of the Project:** Online Voting Survey and Election Opinion Analysis System

❖ **Name of the Company/Client:** Academic Project

❖ **Abstract:**

The project “**Online Voting Survey and Election Opinion Analysis System**” is a web-based application developed to collect and analyse public opinions related to upcoming elections. The system allows citizens to participate in online surveys and express their views on political parties and election-related topics in a secure and user-friendly manner.

The application enables administrators to create surveys, manage questions, and monitor responses in real time. Users can submit their opinions anonymously, ensuring privacy and encouraging honest feedback. The collected survey data is stored in a MongoDB database and processed using analytical techniques to identify voting patterns and public sentiment trends.

The frontend of the system is developed using **Angular**, providing a responsive and interactive user interface. The backend is implemented using **Python FastAPI**, which handles API requests efficiently and ensures high performance. **MongoDB** is used as the database for flexible and scalable data storage.

The main objective of the system is to support data-driven decision-making by predicting possible election outcomes based on public opinion. This project helps researchers, analysts, and organizations understand voter behaviour while promoting digital participation in democratic processes.

❖ **Introduction and Objectives of the Project:**

The **Online Voting Survey and Election Opinion Analysis System** is designed to address the need for a digital platform to collect and analyse public opinions related to elections. In a democratic system, understanding voter sentiment before elections plays an important role in planning, analysis, and decision-making. Traditional opinion surveys are time-consuming, costly, and limited in reach.

This project provides an **online, web-based solution** that enables users to participate in election-related surveys easily from anywhere. The system allows administrators to create surveys and analyse responses efficiently, while users can submit their opinions securely and anonymously. The collected data is stored and processed to identify voting trends and public sentiment.

The application uses **Angular** for the frontend to ensure a responsive and interactive user interface. The backend is developed using **Python FastAPI**, which offers fast and secure RESTful API services. **MongoDB** is used as the database for flexible and scalable data storage. The system aims to promote digital participation, transparency, and data-driven insights in the election analysis process.

Objectives:

1. To design and develop an **online voting survey system** for collecting public opinions related to elections.
2. To provide a **user-friendly and responsive interface** using Angular for easy participation.
3. To develop a **secure and efficient backend** using Python FastAPI for handling survey data.
4. To store and manage survey responses using **MongoDB** for scalability and flexibility.
5. To enable **anonymous and secure submission** of user opinions to encourage honest feedback.
6. To analyse collected data and generate insights to **predict possible election outcomes**.

❖ **Project Category:** Web Application

❖ **Tools / Platform, Hardware and Software Requirement specifications:**

Operating System Platform:

- Windows 10 / 11 or
- Ubuntu 22.04 LTS (Linux Environment)

Hardware Requirements:

- Processor: Intel Core i3 or equivalent
- Hard Disk: Minimum 50 GB
- RAM: 4 GB or above

Software Requirements:

✓ **Front-End (4GL Programming Language):**

- Angular
- HTML5
- CSS3
- JavaScript
- TypeScript

✓ **Middle Layer / Backend Framework:**

- Python 3.x
- FastAPI Framework

✓ **Back End (Database):**

- MongoDB

✓ **Report Generation Tool:**

- Chart.js / Matplotlib (for data visualization)

✓ **Other Tools:**

- Visual Studio Code – Code editor
- Postman – API testing
- Git & GitHub – Version control
- Docker – Application containerization and deployment
- Node.js & npm – Angular package management

❖ **Scope of the Project**

The “**Online Voting Survey and Election Opinion Analysis System**” aims to provide a complete digital platform for collecting, managing, and analysing public opinions related to elections. The system focuses on simplifying the process of conducting opinion surveys, increasing public participation, and generating meaningful insights from collected data.

The project is developed as a **web-based application** with a modular and scalable architecture. It allows users to participate in surveys securely and anonymously, while administrators can manage surveys and analyse results efficiently. The system is designed in such a way that it can be extended in the future for large-scale election analysis and advanced analytics.

In Scope:

1. **User Management:** Secure user registration, login, and role-based access for administrators and users.
2. **Survey Management:** Creation, modification, and management of election-related surveys and questions by the administrator.
3. **Opinion Collection:** Allows users to submit their opinions or votes anonymously through an online interface.
4. **Data Storage:** Storage and retrieval of survey responses using **MongoDB** for flexible and scalable data management.

5. **Admin Dashboard:** Provides administrators with tools to monitor survey participation, view results, and manage users.
6. **RESTful APIs:** Backend APIs developed using **Python FastAPI** to handle survey operations and data processing.
7. **Data Analysis and Reporting:** Basic analysis of collected data and visualization of results in the form of charts and reports.

Out of Scope (Future Enhancements):

1. Integration with **advanced data analytics or AI/ML models** for accurate election prediction.
2. Support for **mobile applications** (Android/iOS).
3. Real-time opinion tracking using **live dashboards**.
4. Integration with government or external election data sources.
5. Multi-language support for wider public participation.

Overall, the system is designed to be secure, scalable, and maintainable, providing a strong foundation for future enhancements in election opinion analysis and digital survey systems.

❖ Proposed system

The proposed system, **Online Voting Survey and Election Opinion Analysis System**, is a modern and digital solution designed to collect, manage, and analyse public opinions related to elections. Unlike traditional offline surveys and manual opinion polls, this system provides an online platform that enables wider participation, faster data collection, and accurate analysis.

The system is developed as a **web-based application** using **Angular** for the frontend, which offers a responsive and user-friendly interface for survey participation. The backend is implemented using **Python FastAPI**, which provides secure, high-performance RESTful APIs to handle survey operations, user requests, and data processing.

User responses and survey data are stored in **MongoDB**, a NoSQL database that supports flexible and scalable data storage. The system allows users to submit their opinions anonymously, ensuring privacy and encouraging honest participation. Administrators can manage surveys, view results, and analyse voting trends through an admin dashboard.

The proposed system focuses on transparency, accessibility, and data-driven insights, making it an effective tool for understanding public sentiment and predicting possible election outcomes.

Key Features of the Proposed System:

1. **User Participation Module:**
Allows users to participate in election-related surveys easily through a web interface.
2. **Survey Management:**
Enables administrators to create, update, and manage surveys and questions.
3. **Secure Backend APIs:**
RESTful APIs developed using Python FastAPI ensure secure and efficient data handling.
4. **Data Storage and Management:**
MongoDB is used to store survey responses, user data, and survey metadata.
5. **Admin Dashboard:**
Provides administrators with access to survey results, response statistics, and participation reports.
6. **Data Analysis and Reporting:**
Basic analytical tools and visualizations are used to identify trends and public opinion patterns.

Advantages of the Proposed System:

- Eliminates the limitations of manual and offline survey methods.
- Enables quick and large-scale collection of public opinions.
- Ensures user privacy through anonymous participation.
- Provides accurate and organized data for analysis and decision-making.
- Scalable architecture that can be enhanced with advanced analytics or mobile applications in the future.

❖ **Module Specification:** [No. of Modules & their Specification]

The **Online Voting Survey and Election Opinion Analysis System** is organized into **four main modules**, each responsible for a specific functionality of the system. The modular architecture improves maintainability, scalability, and ease of future enhancements. The system follows a **web-based client-server model**, where users participate in surveys and administrators manage and analyse election-related data.

1. User Management Module

Purpose:

The User Management Module handles user registration, authentication, and access control within the system.

Specifications:

- Manages user registration and login functionality.
- Supports **role-based access** for Admin and Users.
- Ensures secure access to the system using backend API validations.
- Allows anonymous participation for users while restricting administrative actions.
- Maintains user profile information and participation records.
- Prevents duplicate or unauthorized access to surveys.

2. Survey Management Module**Purpose:**

The Survey Management Module is responsible for creating, managing, and controlling election-related surveys and questions.

Specifications:

- Allows administrators to create, update, and delete surveys.
- Supports multiple question types such as single-choice and multiple-choice.
- Enables setting survey start and end dates.
- Manages survey visibility and participation rules.
- Ensures each user can submit responses according to defined rules.
- Stores survey metadata and configurations in MongoDB.

3. Opinion Collection and Analysis Module**Purpose:**

This module collects user responses and analyses survey data to identify voting trends and public opinion patterns.

Specifications:

- Collects survey responses submitted by users.
- Stores responses securely in MongoDB.
- Vote count per option
- Percentage distribution
- Trend identification
- Generates analytical summaries for administrators.
- Prepares data for visualization and reporting.

4. Reporting and Dashboard Module

Purpose:

The Reporting and Dashboard Module provides a centralized interface to view survey results and system performance.

Specifications:

- Displays survey results in the form of charts, graphs, and tables.
- Provides real-time statistics such as total participants and response rates.
- Active and completed surveys
- Opinion trends and summary reports

- Helps decision-makers interpret public sentiment effectively.
- Supports export of reports for further analysis (future enhancement).

❖ **Bibliography:**

- **Angular Framework Documentation**
<https://angular.io/docs>
- **Node.js Official Documentation**
<https://nodejs.org/en/docs>
- **Python Official Documentation**
<https://docs.python.org/3/>
- **FastAPI Framework Documentation**
<https://fastapi.tiangolo.com/>
- **MongoDB Official Documentation**
<https://www.mongodb.com/docs/>
- **Docker Documentation**
<https://docs.docker.com/>
- **Visual Studio Code Documentation**
<https://code.visualstudio.com/docs>
- **Postman API Documentation**
<https://learning.postman.com/docs/>
- **Git and GitHub Documentation**
<https://docs.github.com/>
- **Project Source Code (GitHub Repository)**
<https://github.com/UdhayarajanJ/2202561-MCA-2025-2027.git>

Student Email: udhayarajanj@gmail.com

Student Contact No: 8248748016

Date:

Place: Pune

Signature of Student